

Arjuna JEE 2025

Maths

Sequence & Series

DPP: 7

Q1 If $a_1, a_2, a_3, \dots, a_n$ are in AP, where $a_i > 0$ for all i , then the value of

$$\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}}$$
 is

- (A) $\frac{n-1}{\sqrt{a_1} + \sqrt{a_n}}$
 (B) $\frac{n+1}{\sqrt{a_1} + \sqrt{a_n}}$
 (C) $\frac{n-1}{\sqrt{a_1} - \sqrt{a_n}}$
 (D) $\frac{n+1}{\sqrt{a_1} - \sqrt{a_n}}$

Q2 The sum of the series $1.2.3 + 2.3.4 + 3.4.5 + \dots$ to n terms is

- (A) $n(n+1)(n+2)$
 (B) $(n+1)(n+2)(n+3)$
 (C) $\frac{1}{4}n(n+1)(n+2)(n+3)$
 (D) $\frac{1}{4}(n+1)(n+2)(n-3)$

Q3 The sum of n terms of the series

$$\frac{3}{1^2} + \frac{5}{1^2+2^2} + \frac{7}{1^2+2^2+3^2} + \dots$$
 is

- (A) $\frac{n+1}{n}$
 (B) $\frac{n}{n+1}$
 (C) $\frac{6n}{n+1}$
 (D) $\frac{6(n-1)}{n}$

Q4 $\frac{3}{1^2 2^2} + \frac{5}{2^2 3^2} + \frac{7}{3^2 4^2} + \dots$ upto n terms is equal to

- (A) $1 - \frac{1}{(n+1)^2}$
 (B) $1 - \frac{1}{n^2}$
 (C) $1 - \frac{1}{n(n+1)}$
 (D) $\frac{n}{(n+1)^2}$

Q5

The sum of n terms of the series

$$= \frac{1}{1+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{7}} + \dots$$
 is

- (A) $\sqrt{2n+1}$
 (B) $\frac{1}{2}\sqrt{2n+1}$
 (C) $\sqrt{2n+1} - 1$
 (D) $\frac{1}{2}(\sqrt{2n+1} - 1)$

Q6 The n^{th} term of the series

$$3.8 + 6.11 + 9.14 + 12.17 + \dots$$
 will be

- (A) $3n(n+5)$
 (B) $3n(3n+5)$
 (C) $n(3n+5)$
 (D) $n(n+5)$

Q7 The n^{th} term of series $\frac{1}{1} + \frac{1+2}{2} + \frac{1+2+3}{3} + \dots$ will be

- (A) $\frac{n+1}{2}$
 (B) $\frac{n-1}{2}$
 (C) $\frac{n^2+1}{2}$
 (D) $\frac{n^2-1}{2}$

Q8 If $S = \sum_{r=1}^{80} \frac{r}{(r^4+r^2+1)}$, then the value of $\frac{6481S}{1000}$ is

- (A) 3.2
 (B) 3.24
 (C) 3.26
 (D) 3.28

Q9 The value of

$$\frac{1}{(1+a)(2+a)} + \frac{1}{(2+a)(3+a)} + \frac{1}{(3+a)(4+a)} + \dots + \infty$$
 is, (where, a is a constant)

- (A) $\frac{1}{1+a}$



- (B) $\frac{2}{1+a}$
 (C) ∞
 (D) None of these

Q10 The sum of series $3 + 7 + 13 + 21 + 31 + \dots$ to n terms is equal to

- (A) $\frac{n(n^2+2n+3)}{3}$
 (B) $\frac{n(n^2+3n+5)}{3}$
 (C) $\frac{n(n^2+2n+5)}{3}$
 (D) $\frac{n(n^2+2n+2)}{3}$

Q11 If the sum to n terms of series

$$\frac{1}{1.3.5.7} + \frac{1}{3.5.7.9} + \frac{1}{5.7.9.11} + \dots \text{ is } \frac{1}{90} - \frac{\lambda}{f(n)}$$

then value of $f(\lambda)$ is.

- (A) $640/27$
 (B) $540/27$
 (C) $580/27$
 (D) $620/47$

Q12 If $S_n = \frac{1}{6.11} + \frac{1}{11.16} + \frac{1}{16.21} + \dots$ Upto n terms, then $6S_n$ equal

- (A) $\frac{5n-4}{5n+6}$
 (B) $\frac{n}{5n+6}$
 (C) $\frac{2n-1}{5n+6}$
 (D) $\frac{1}{5n+6}$

Q13 If $\sum_{n=1}^{360} \left(\frac{1}{n\sqrt{n+1} + (n+1)\sqrt{n}} \right) = \frac{p}{q}$ (where p and q relatively prime integers), then $|p - q|$ is

- (A) 0
 (B) 1
 (C) 2
 (D) 3

Q14 If $a_1, a_2, a_3, \dots, a_n$ are in A, P where $a_i > 0$ for all i , then value of

$$\frac{1}{\sqrt{a_1} + \sqrt{a_2}} + \frac{1}{\sqrt{a_2} + \sqrt{a_3}} + \dots + \frac{1}{\sqrt{a_{n-1}} + \sqrt{a_n}}$$

- is equal
 (A) $\frac{n-1}{\sqrt{a_1} + \sqrt{a_n}}$
 (B) $\frac{n+1}{\sqrt{a_1} + \sqrt{a_n}}$

- (C) $\frac{n-1}{\sqrt{a_1} - \sqrt{a_n}}$
 (D) $\frac{n+1}{\sqrt{a_1} - \sqrt{a_n}}$

Q15 If $\sum_{r=1}^{\infty} \frac{1}{(2r-1)^2} = \frac{\pi^2}{8}$, then $\sum_{r=1}^{\infty} \frac{1}{r^2}$ is equal to

- (A) $\frac{\pi^2}{24}$
 (B) $\frac{\pi^2}{3}$
 (C) $\frac{\pi^2}{6}$
 (D) None of these

Q16 If

$$\sum_{k=1}^n k(k+1)(k-1) = pn^4 + qn^3 + tn^2 + sn, \text{ where } p, q, t \text{ and } s \text{ are constants, then the value of } s \text{ is equal to}$$

- (A) $-\frac{1}{4}$
 (B) $-\frac{1}{2}$
 (C) $\frac{1}{2}$
 (D) $\frac{1}{4}$

Q17 The sum of first n terms of the sequence

$$1, 1+2, (1+2+2^2), \dots$$

$(1+2+2^2+\dots+2^{k-1}),$ is of the form $2^{n+R} + S^2 + T_n + U, \forall n \in N$. The value of $(R + S + T + U)$

- (A) -1
 (B) 0
 (C) 1
 (D) -2

Q18 If w is an imaginary cube root of unity then the value of the expression

$$2(1+w)(1+w^2) + 3(2w+1)(2w^2+1) + \dots$$

$$+ \dots + (n+1)(nw+1)(nw^2+1) \text{ is}$$

- (A) $n^2 \left(\frac{n+1}{4} \right)^2$
 (B) $\left(\frac{n(n+1)}{2} \right)^2 + n$
 (C) $\left(\frac{n(n+1)}{2} \right)^2 - n$
 (D) None of these



Answer Key

Q1 (A)

Q2 (C)

Q3 (C)

Q4 (A)

Q5 (D)

Q6 (B)

Q7 (A)

Q8 (B)

Q9 (A)

Q10 (B)

Q11 (A)

Q12 (B)

Q13 (B)

Q14 (A)

Q15 (C)

Q16 (B)

Q17 (D)

Q18 (B)

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